Module 1: The foundation

Cloud setup & access

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Prerequisites

- **Goal:** A secure environment before deploying code.
- **Providers:** DigitalOcean, AWS EC2, Linode, or Raspberry Pi.

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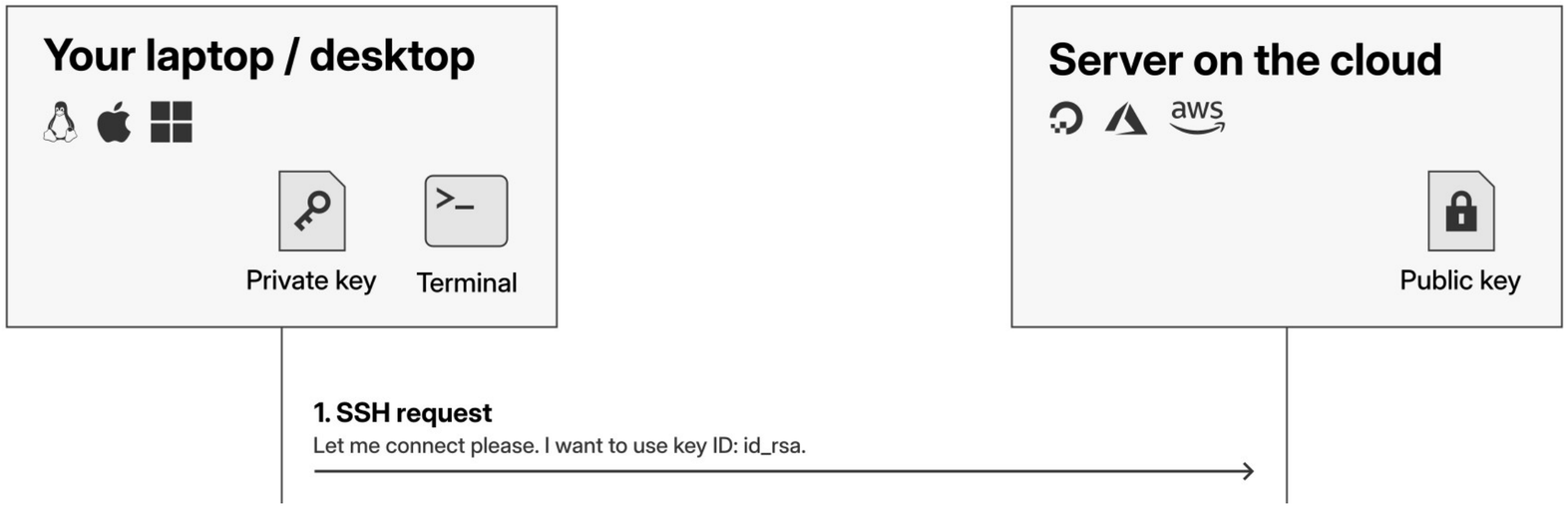
Learning objectives

- Deploy a server running **Ubuntu 24.04 LTS**.
- Generate and configure **SSH Keys** (Ed25519).
- Understand the **"Lock & Key"** security model.
- **Security:** Disable password authentication completely.

How SSH keys work

How SSH keys work



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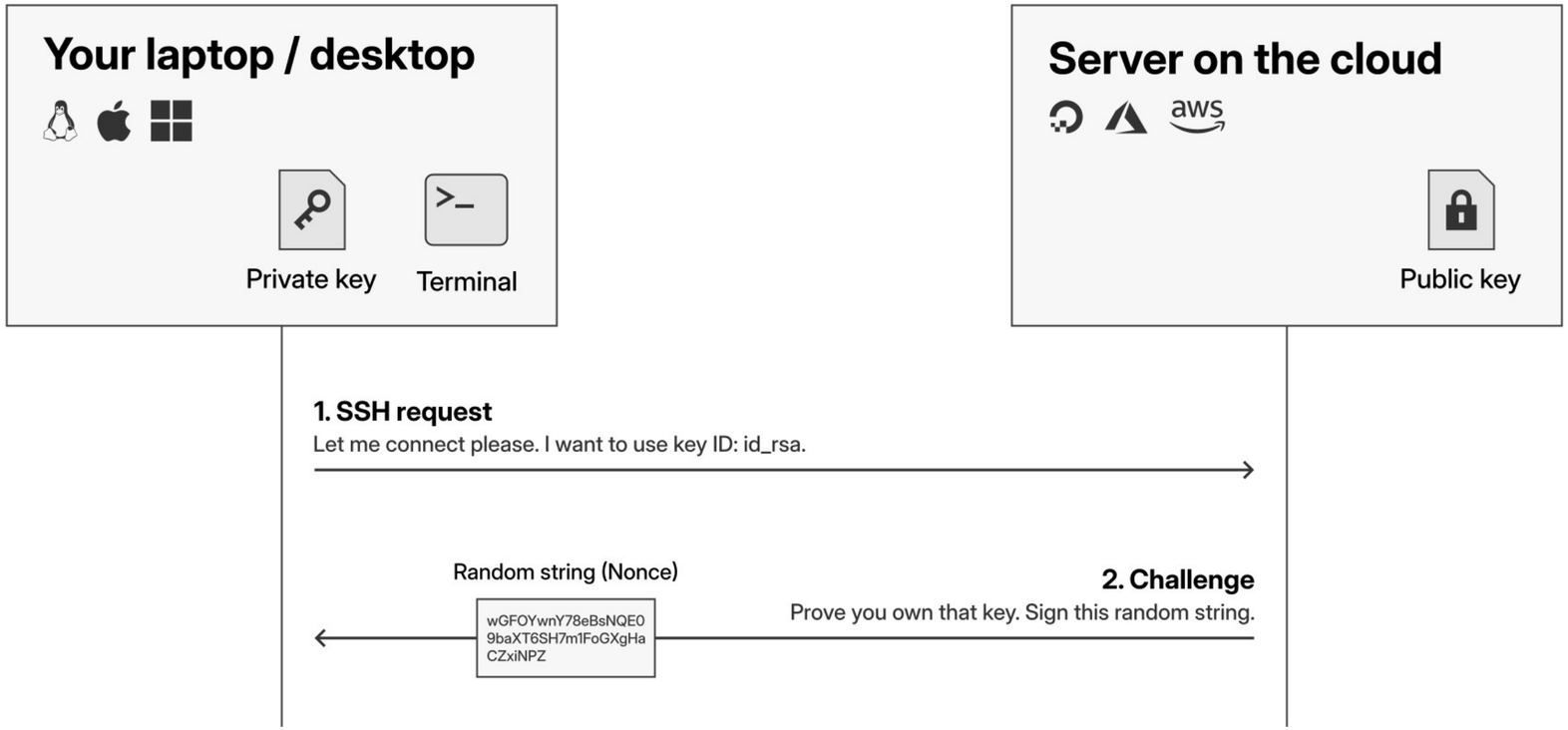

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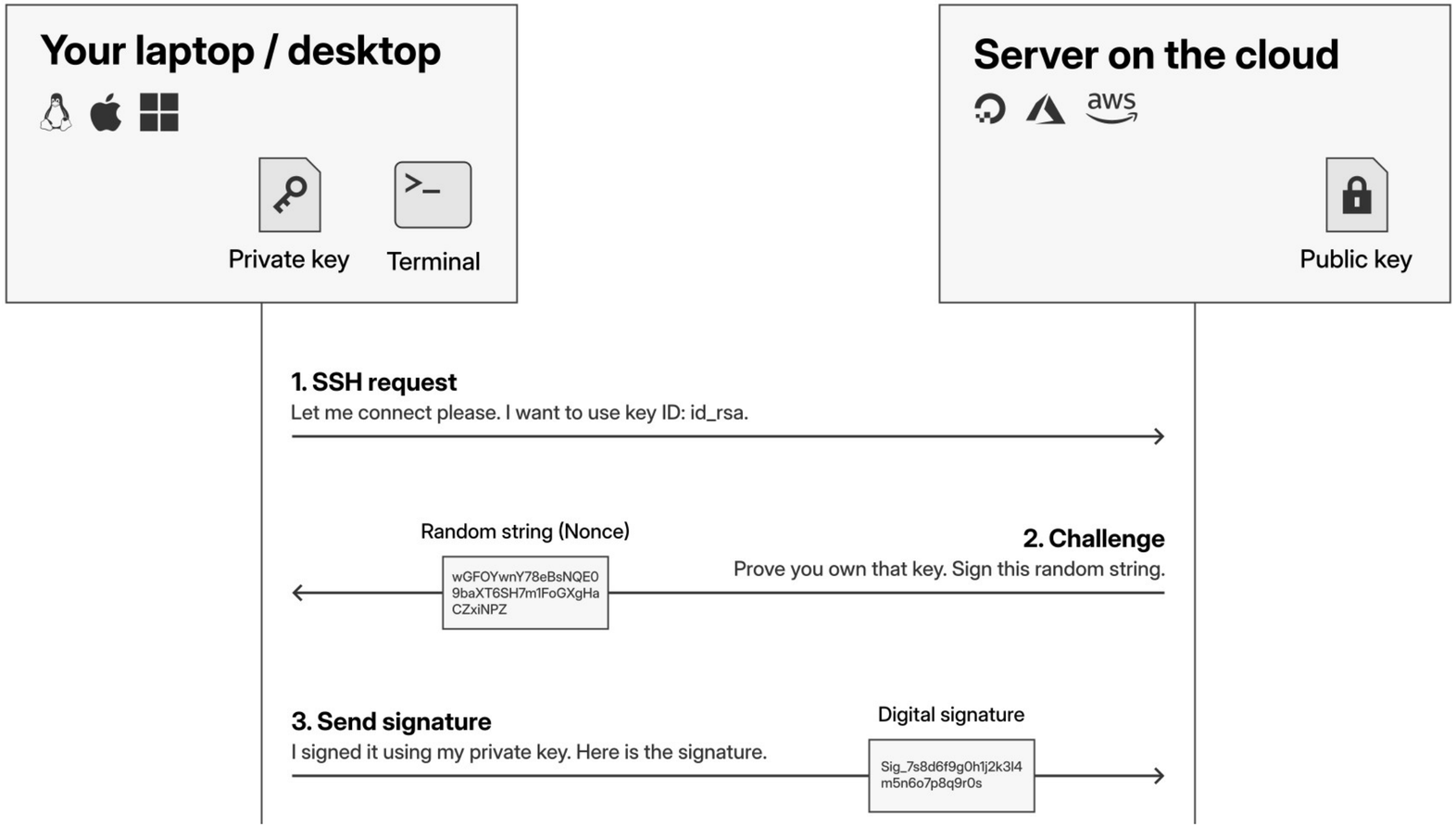

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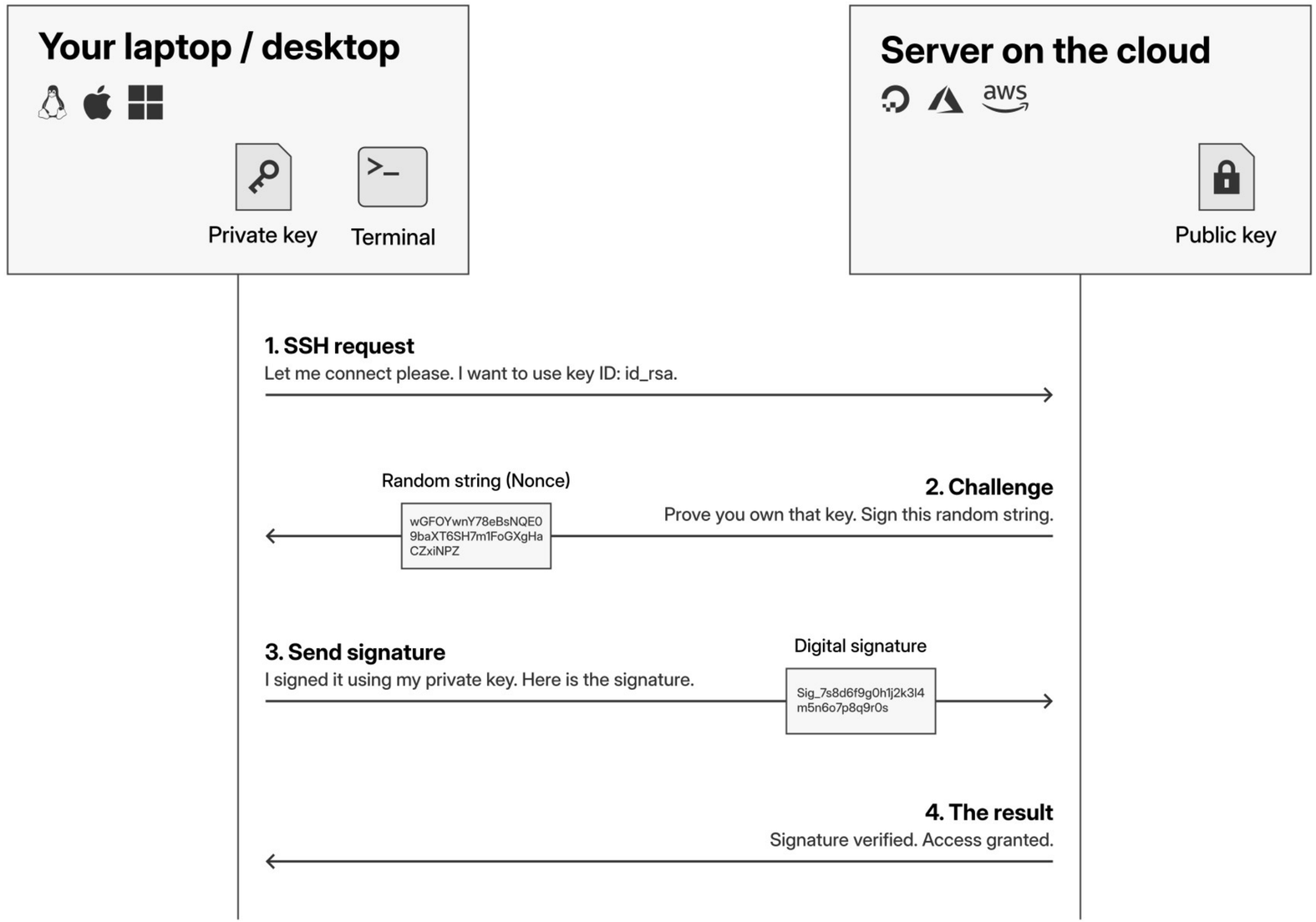


What is next?

- **Current status:** You now have a clean, secure Ubuntu server.
- **The risk:** The server is still exposed to the open internet.
- **Next steps:**
  - **Firewall configuration:** Setting up UFW (Uncomplicated Firewall).
  - **Brute force protection:** Installing Fail2Ban to block attackers.
  - **Emergency access:** How to use the recovery console.

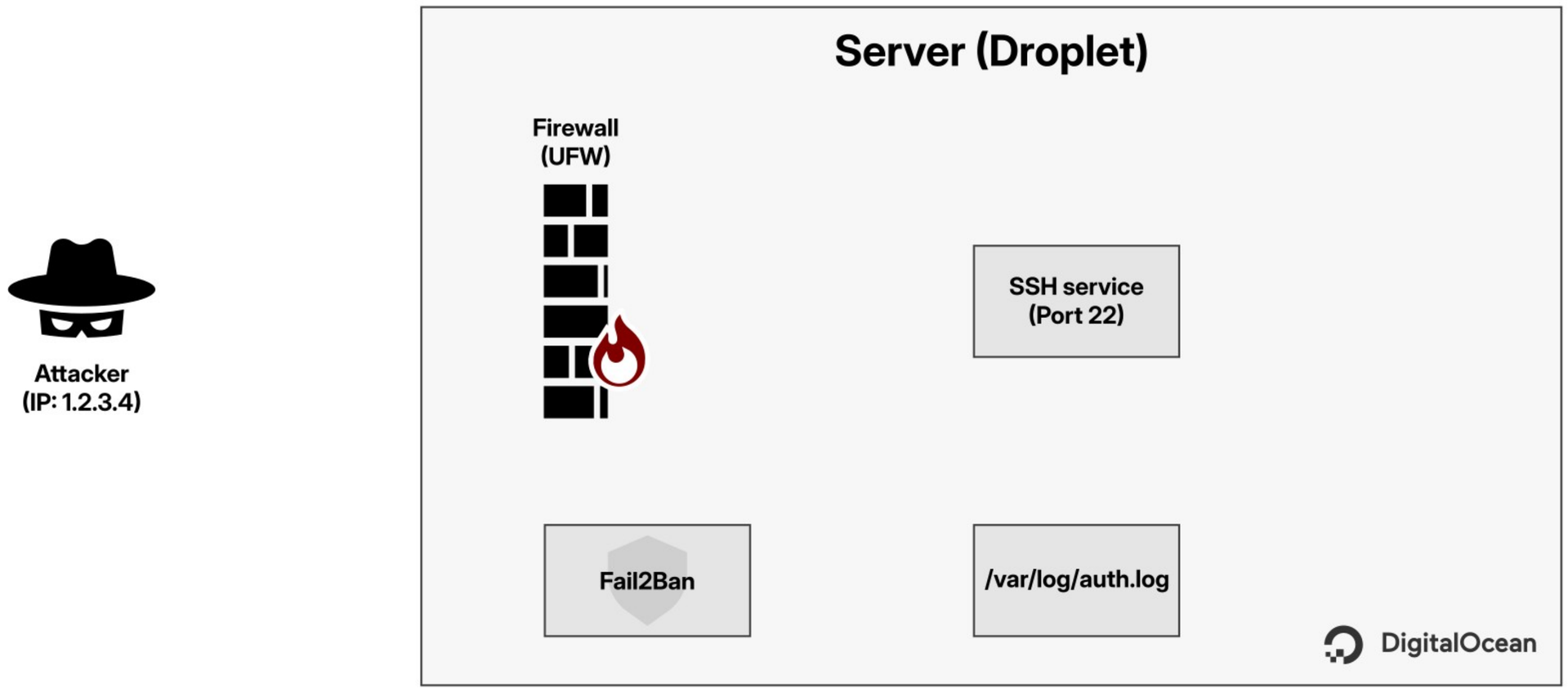
Module 1: The foundation

The firewall strategy

The firewall strategy

- **Problem:** Default Linux servers accept traffic on all ports.
- **Solution:** Use UFW to block everything by default.
- **Automation:** Use Fail2Ban to block bots instantly.
- **Safety net:** Using the DigitalOcean Recovery Console.

How Fail2Ban works



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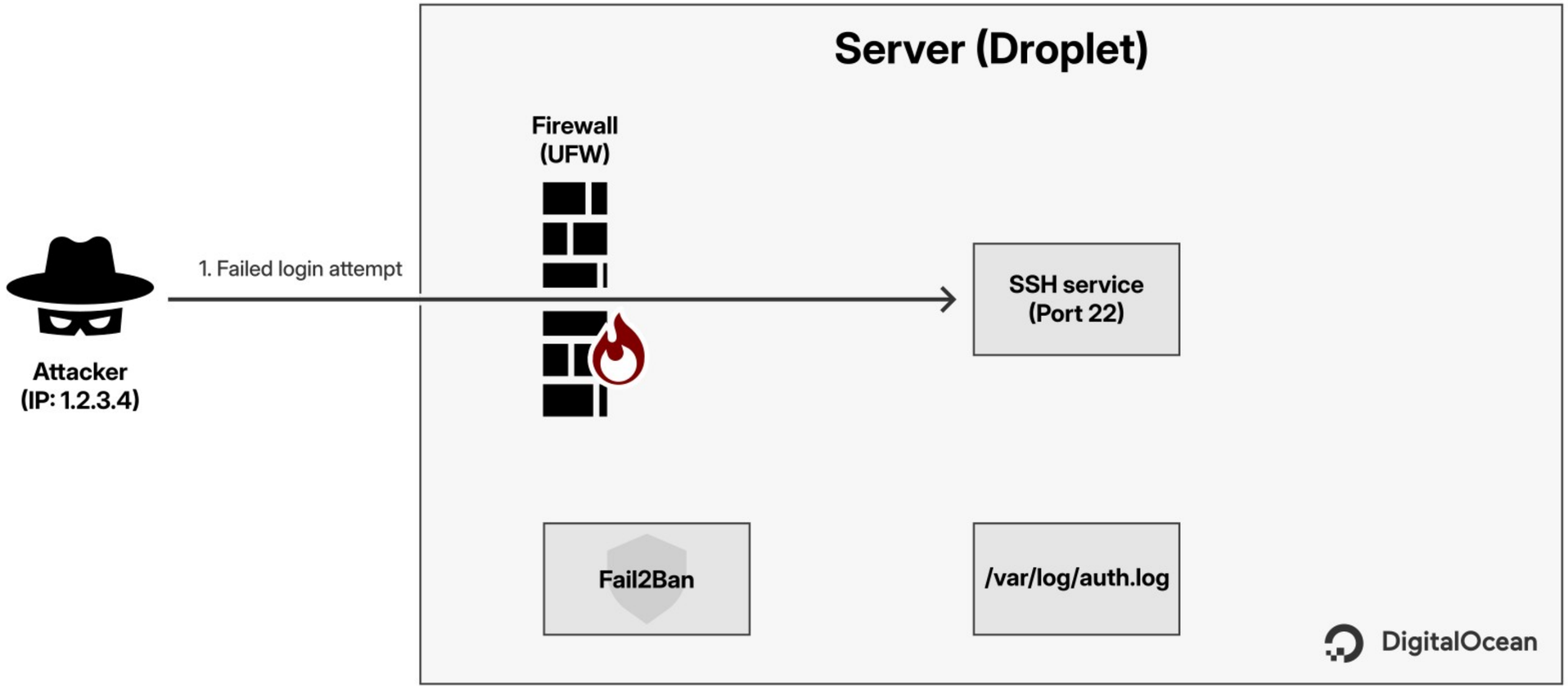

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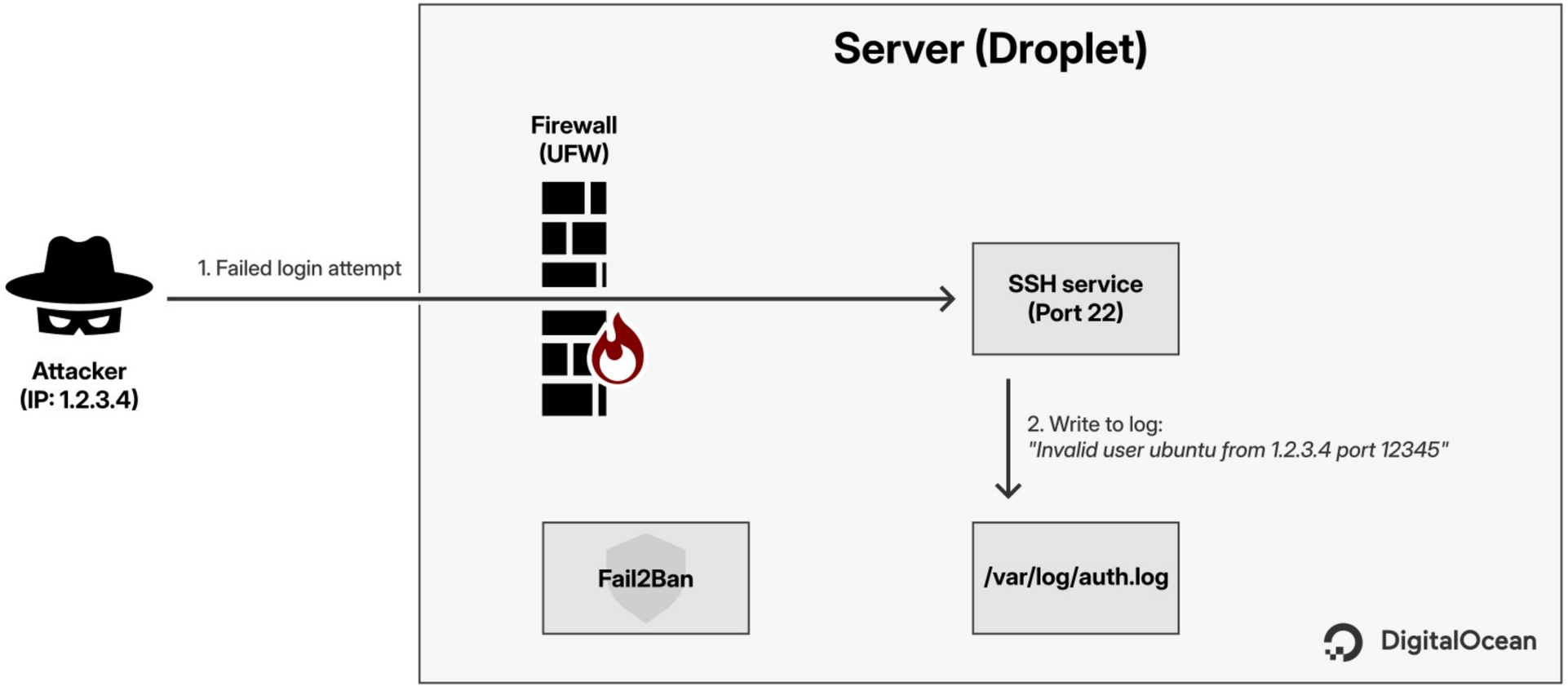

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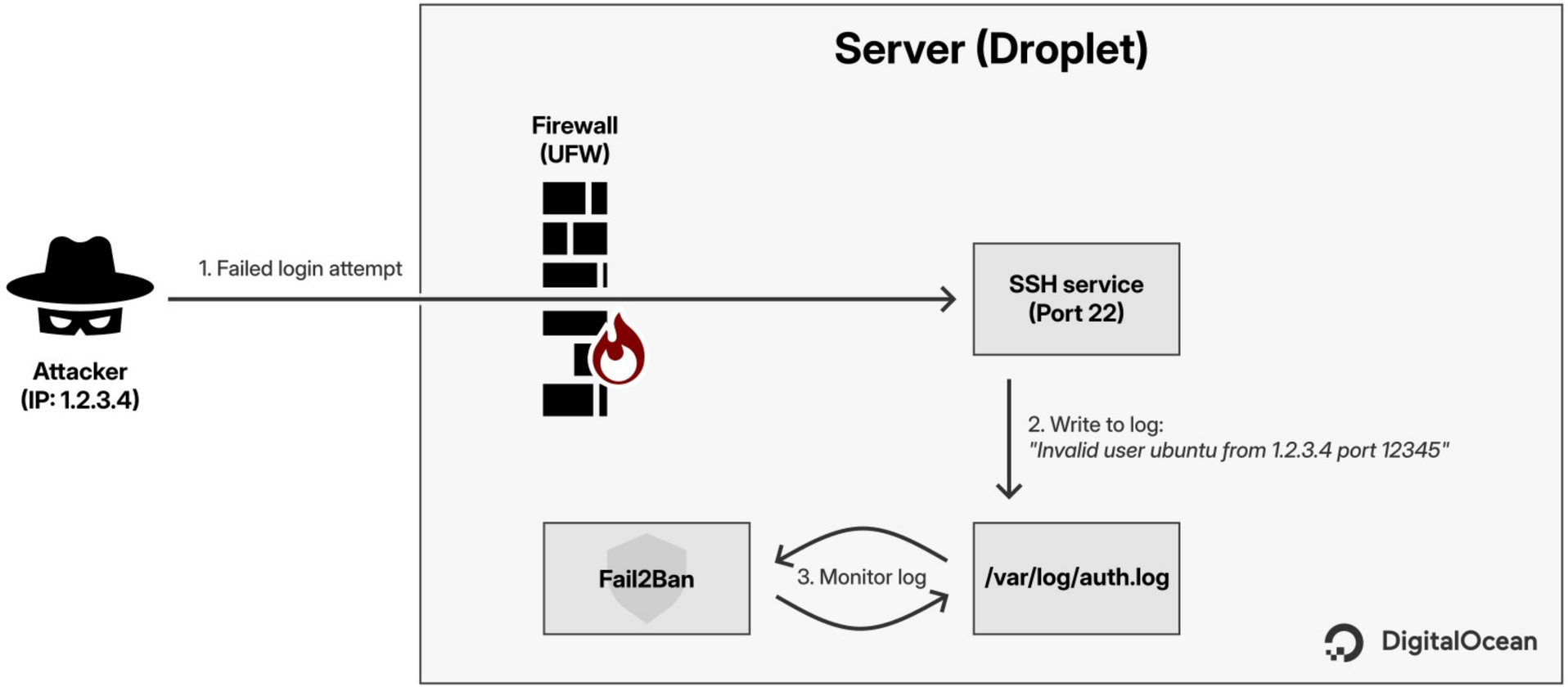

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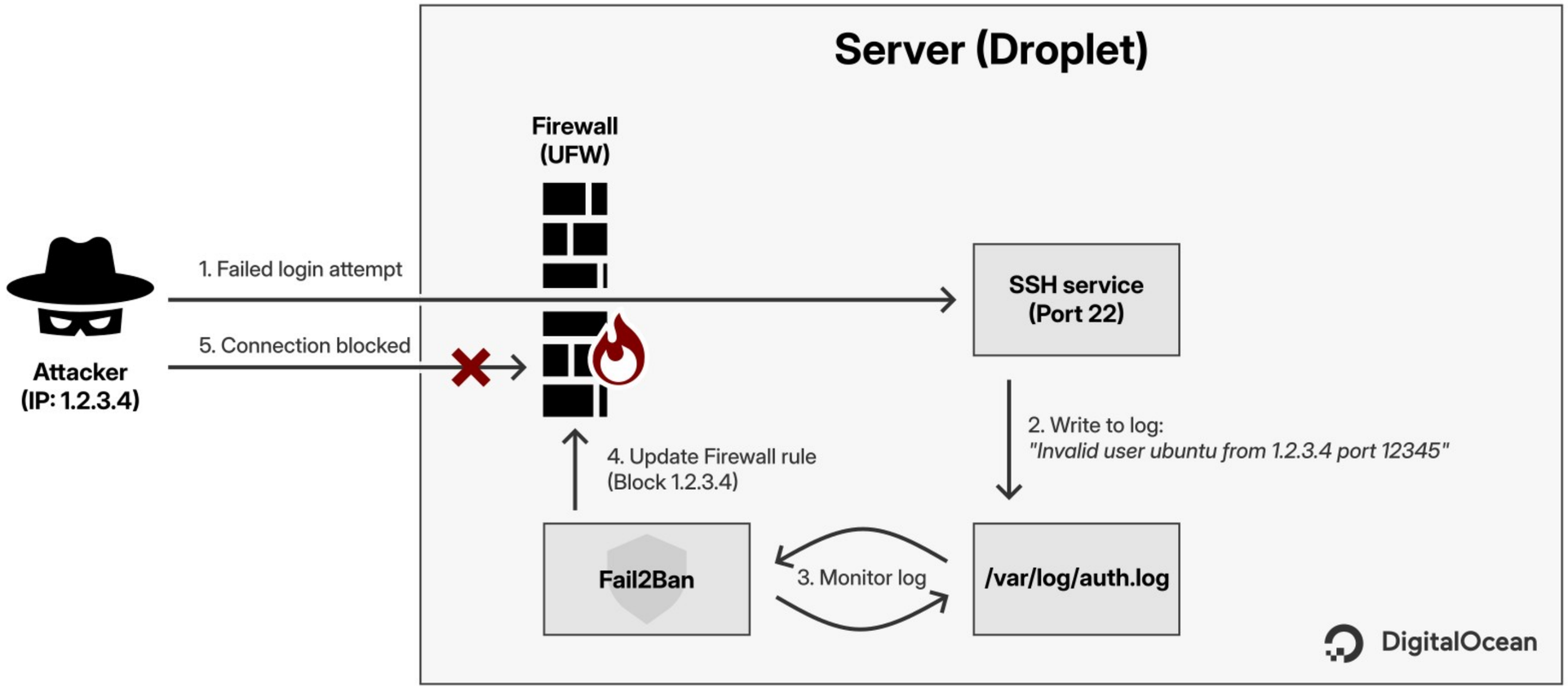
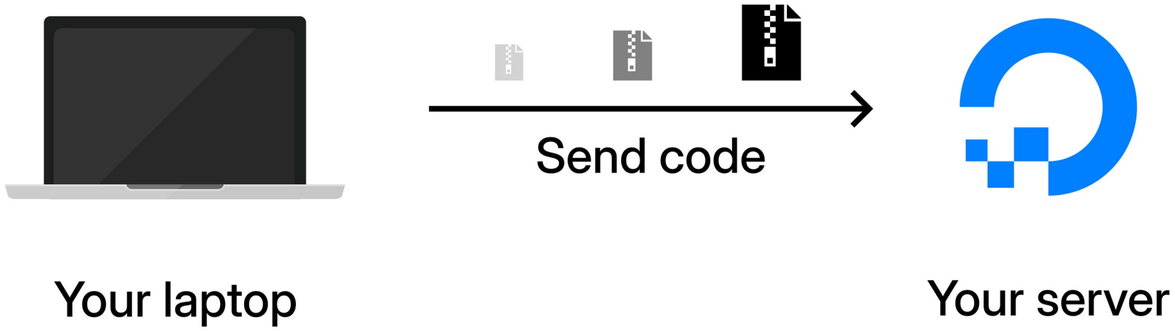


Module 2: The application runtime

Manual deployment

Manual deployment

- Move code using **scp** and organize permissions.



Manual deployment

- Move code using **scp** and organize permissions.
- Install **Python** (Backend) and **Node.js** (Frontend).
- Fix build errors using **Swap files**.
- Use **SSH Tunnels** to bypass the **firewall**.

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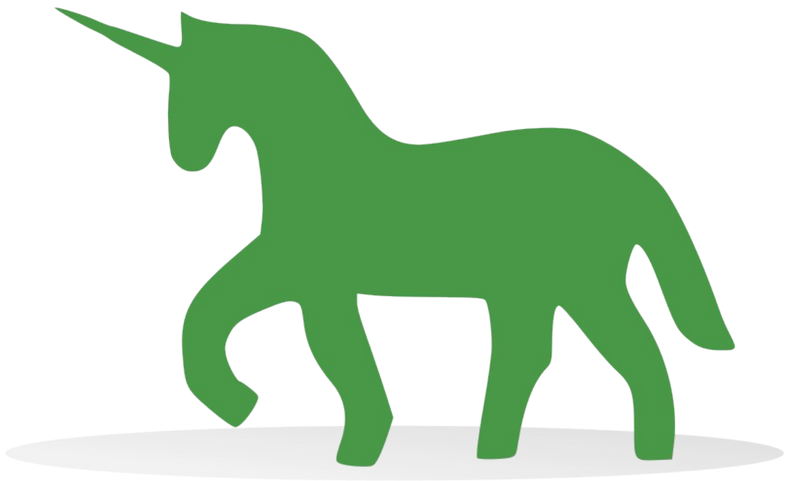
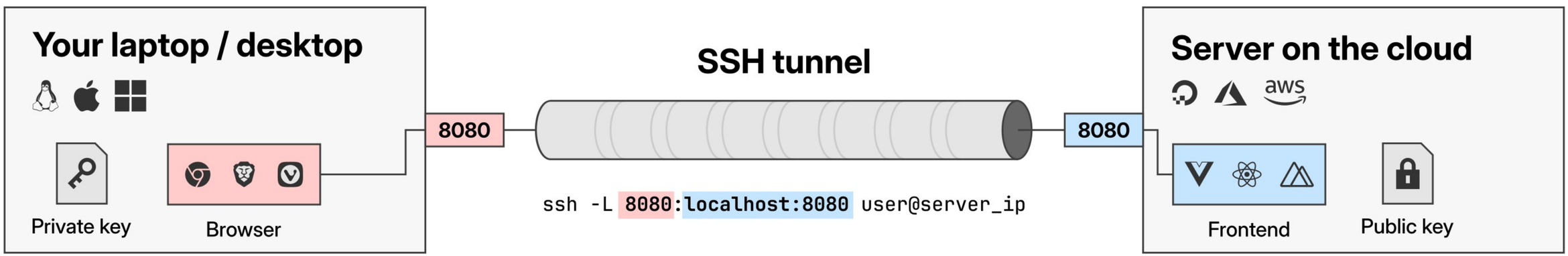
- Configure **Gunicorn** and **Supervisor**.



SSH tunneling



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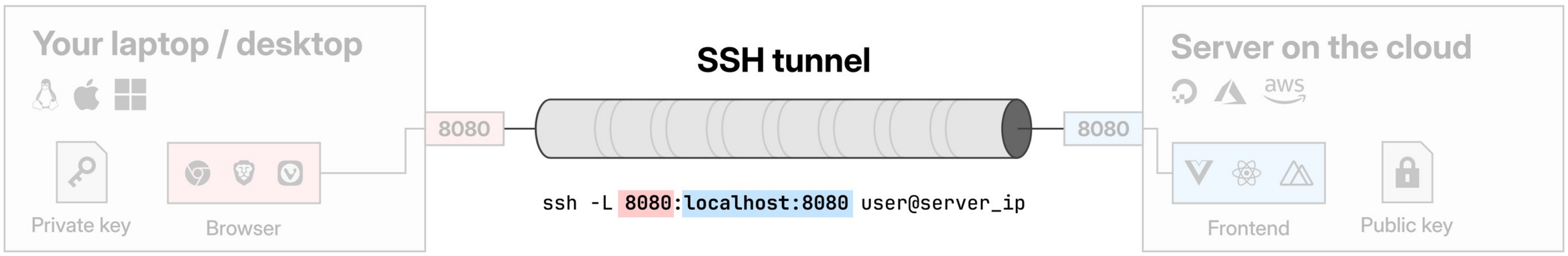

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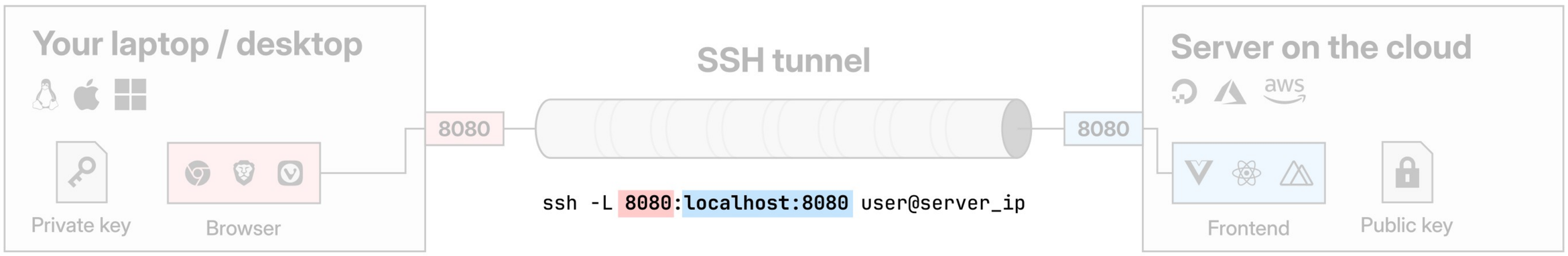

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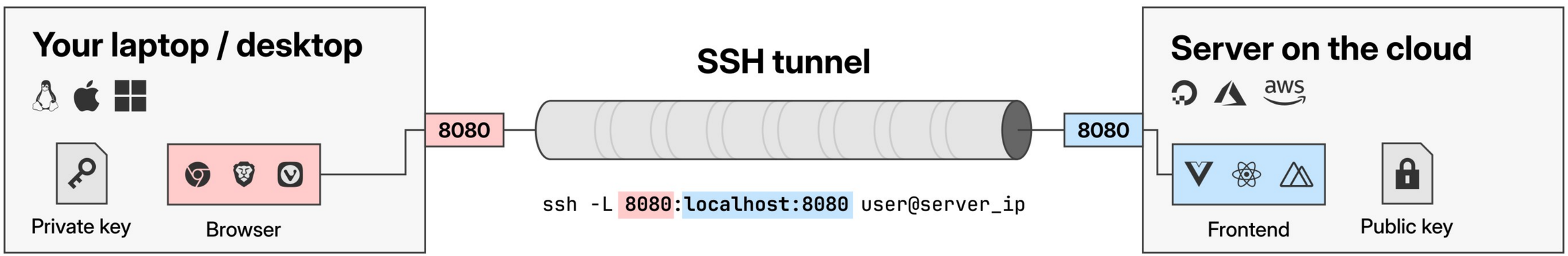
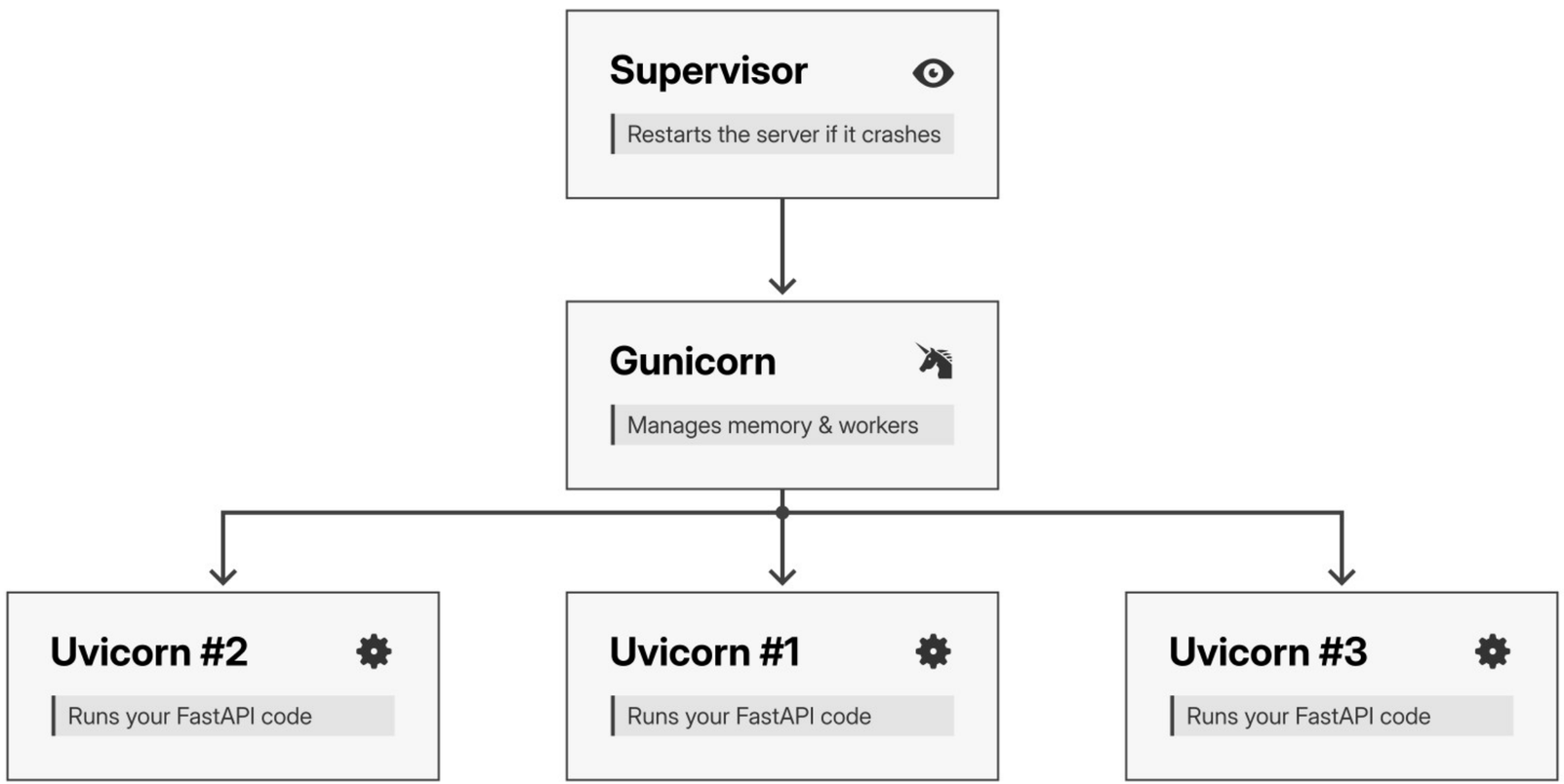


Keeping the app alive



Module 2: The application runtime

Reverse proxy & security headers

Reverse proxy & Security

- Sync frontend and backend URLs.
- Install and configure **Nginx**.

Reverse proxy & Security

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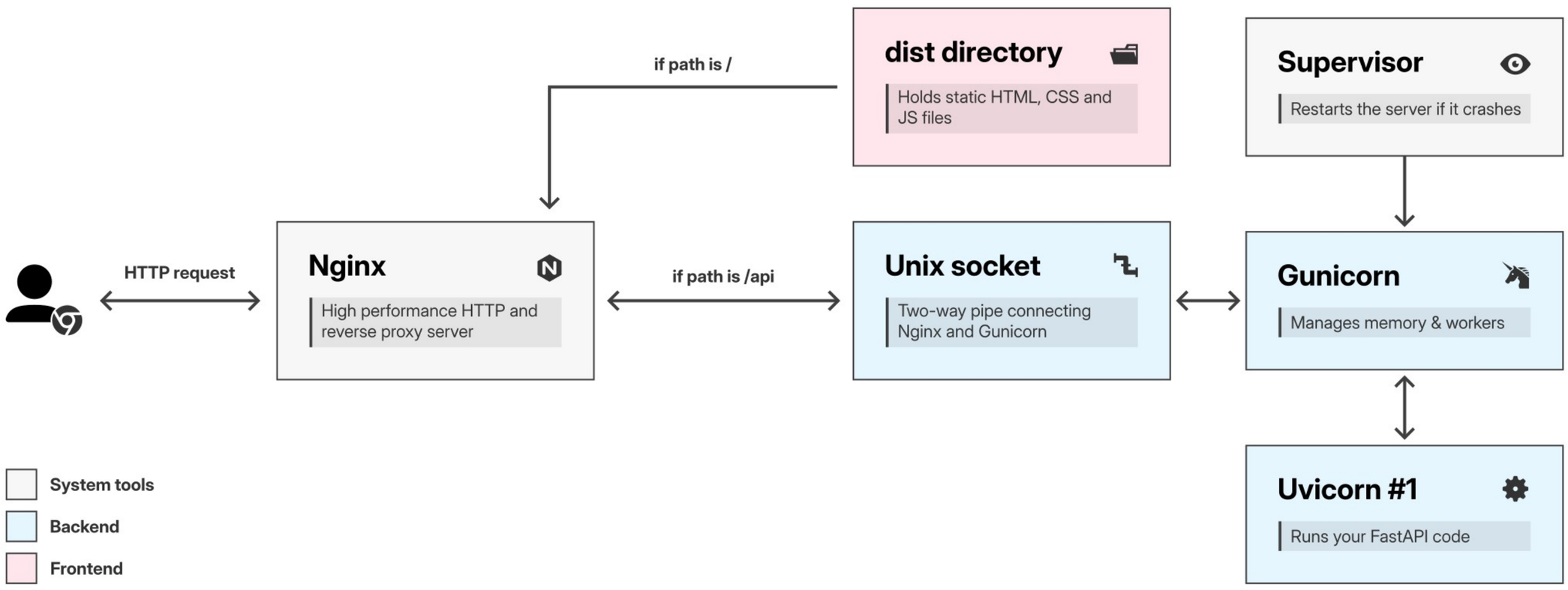
- Understand and fix the **502 Bad Gateway** error.

Reverse proxy & Security

- Sync frontend and backend URLs.
- Install and configure **Nginx**.
- Understand and fix the **502 Bad Gateway** error.
- Protect users with **HTTP Security Headers (CSP)**.

Nginx: The traffic cop



Module 3: Data and search

Self-hosting Meilisearch

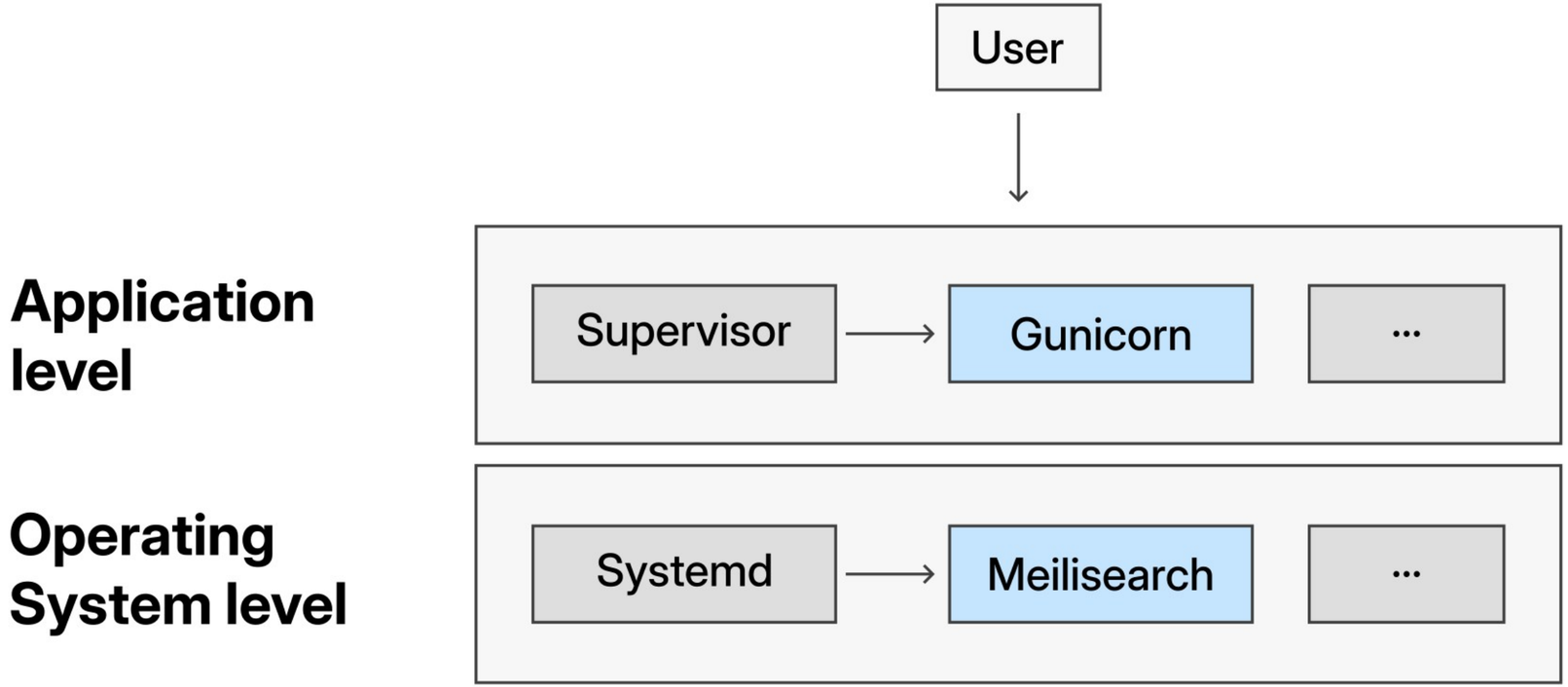
Self-hosting a search engine

- **The engine**: Meilisearch (a lightning-fast Rust binary).
- **Security**: Isolate the database using a **system user**.
- **Automation**: Run it as a background service using **systemd**.
- **Maintenance**: Schedule automatic snapshot backups.
- **Management**: Use an **SSH Tunnel** to access a visual dashboard securely.

What is a system user?

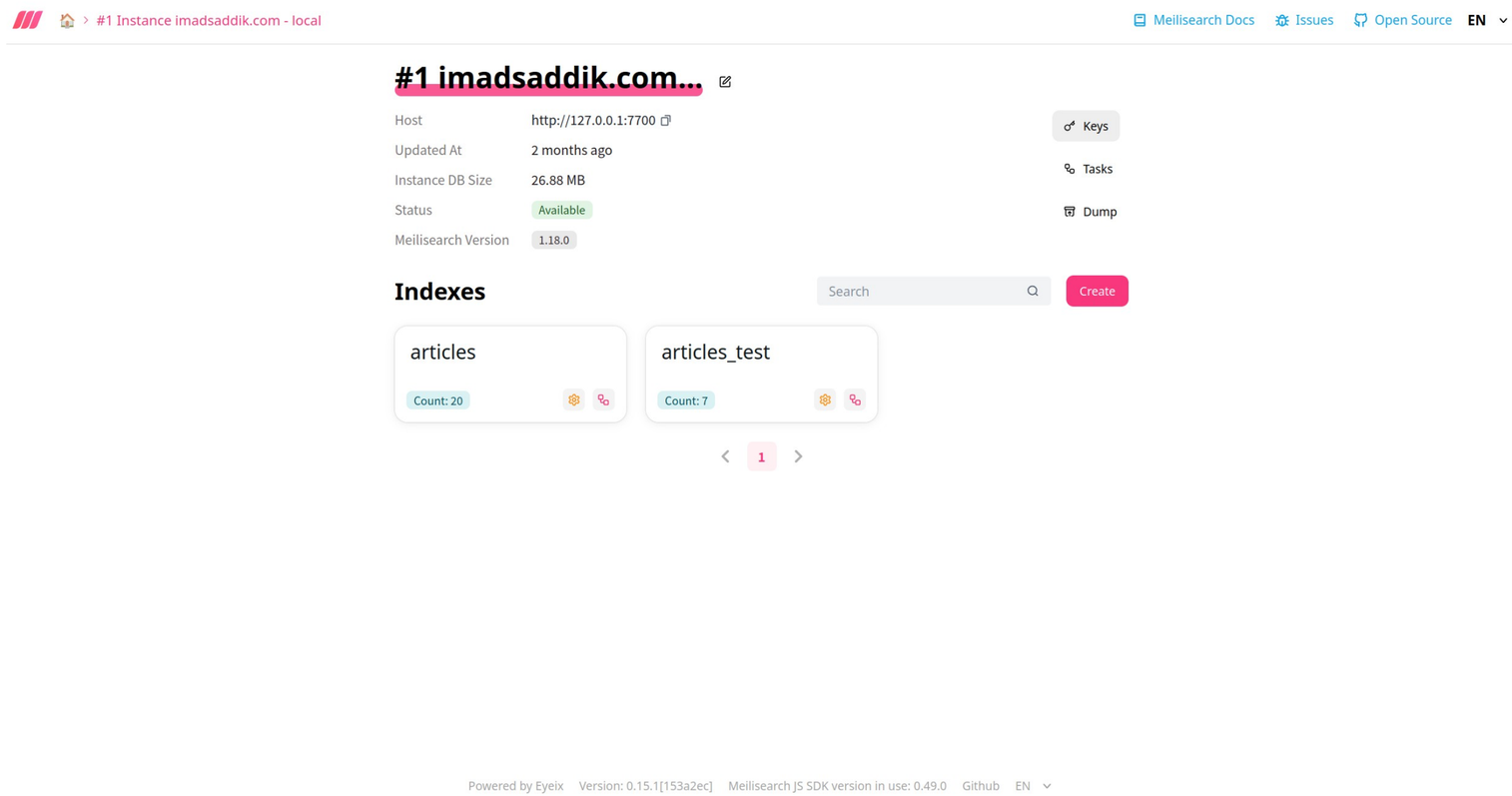
Feature	Standard user	System user
Purpose	Human administration	Running background services
Shell	/bin/bash	/bin/false
Login	Yes	No
Home folder	/home/user	/var/lib/meilisearch

Systemd vs Supervisor



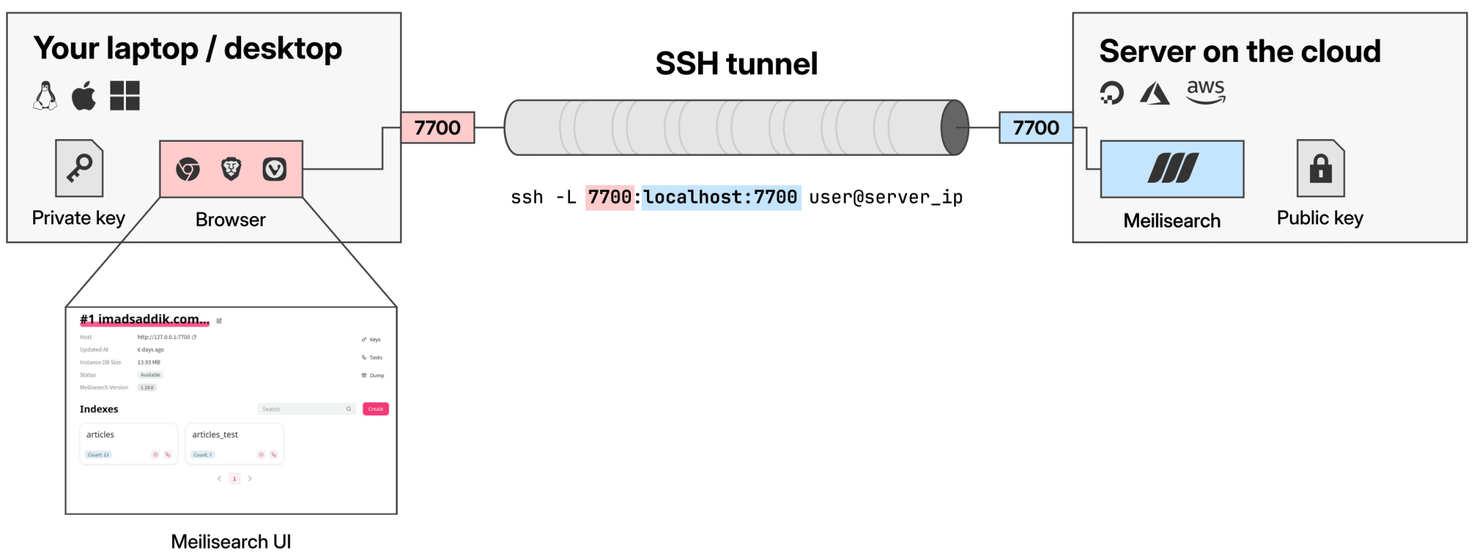
Secure database management

- **The tool**: meilisearch-ui (Open-source web dashboard).
- **Features**: Search, add, and edit documents visually.



Secure database management

- **The tool**: meilisearch-ui (Open-source web dashboard).
- **Features**: Search, add, and edit documents visually.
- **The connection**: Secure access via SSH Tunneling (no open ports).



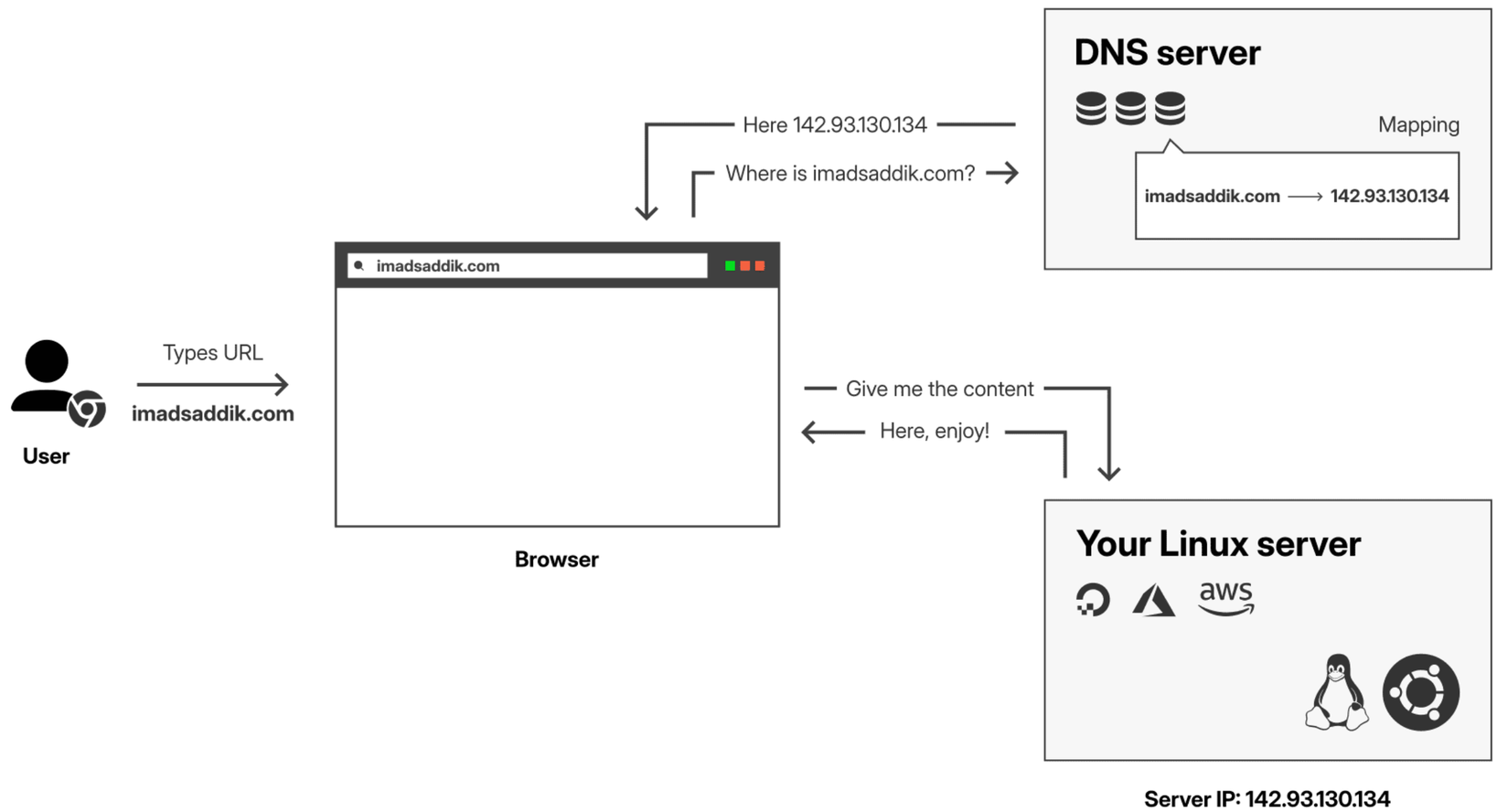
Module 4: Global delivery and security

Domains and SSL

Domains & SSL

- Purchase a **custom domain**.
- Configure **DNS Records** (A and CNAME).
- Install free **SSL certificates** via Let's Encrypt.
- Automate 90-day **renewals** using Cron.

How DNS works

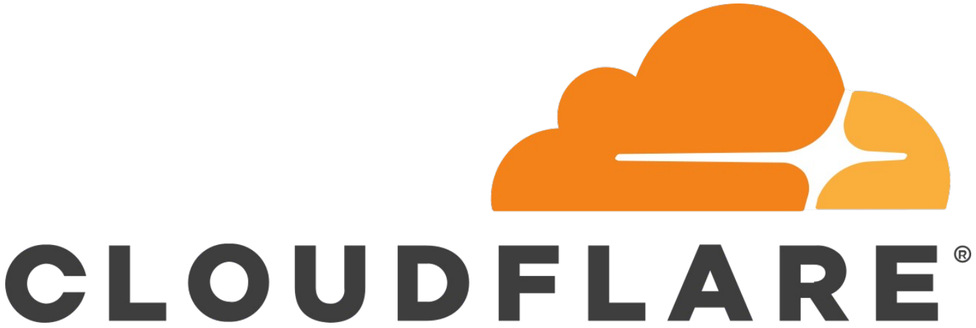


Module 4: Global delivery and security

The CDN layer and caching

Global speed & Caching

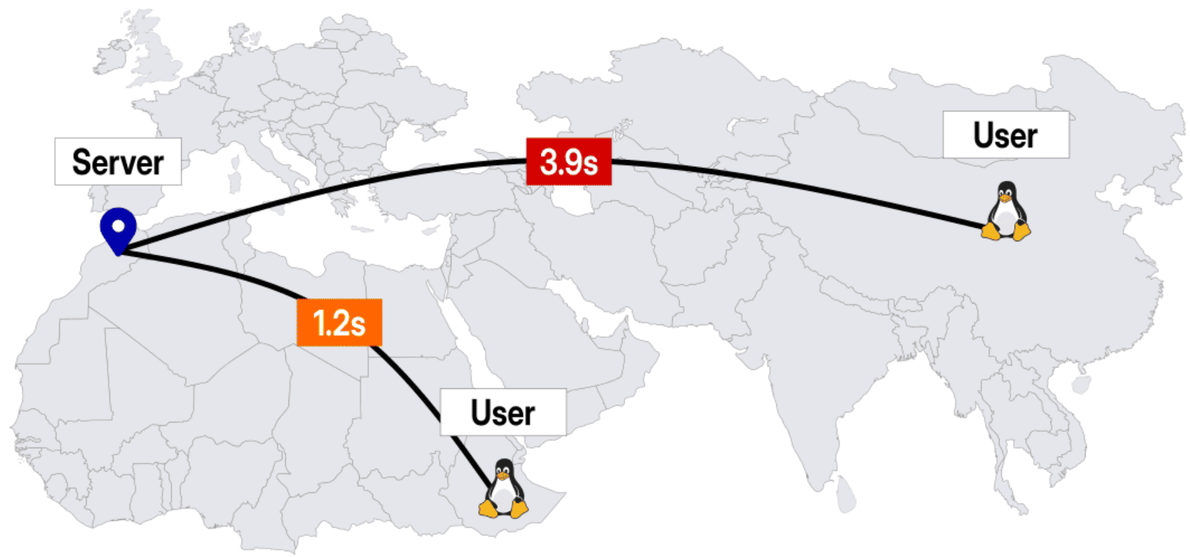
- Route traffic through a **Content Delivery Network (CDN)**.
- Configure strict **end-to-end SSL encryption**.
- Unmask visitor IP addresses in Nginx.
- Implement a **caching strategy** for SPAs.



Beating the speed of light

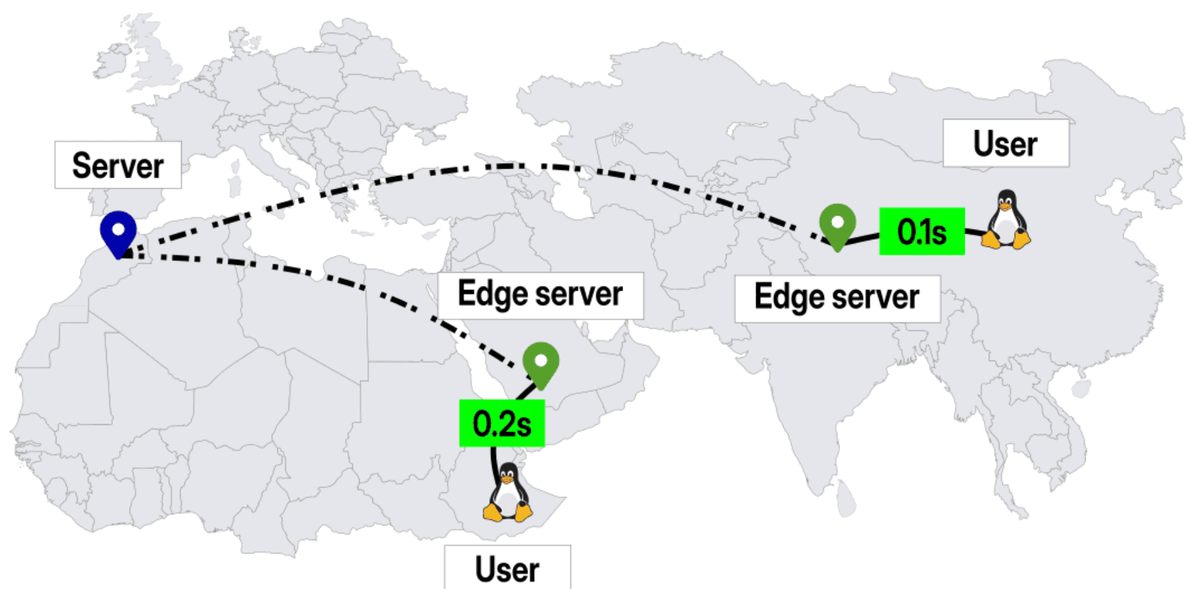
Before CDN

Users connect directly to your origin server. Physical distance causes slow load times for international visitors.



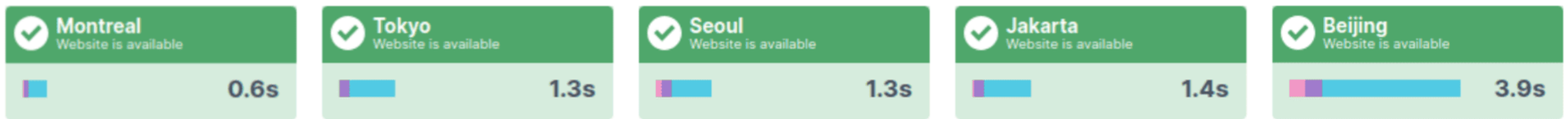
After CDN

A global network of edge servers caches your files. Users connect to a server right next to them, resulting in lightning-fast load times.

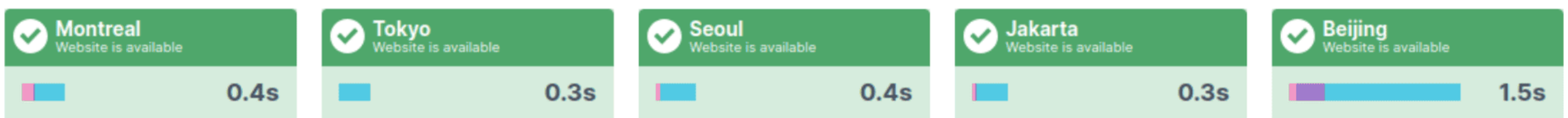


Beating the speed of light

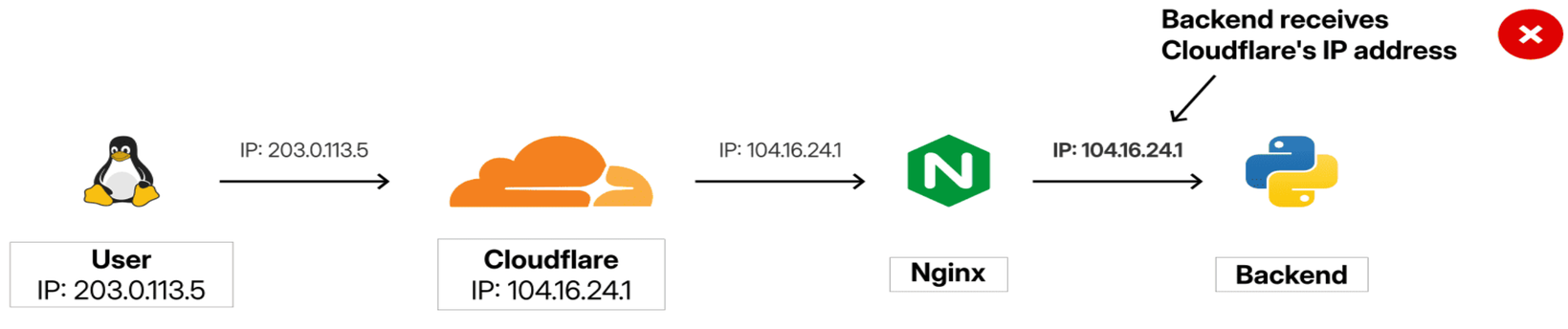
Before using Cloudflare



After using Cloudflare



The real IP problem



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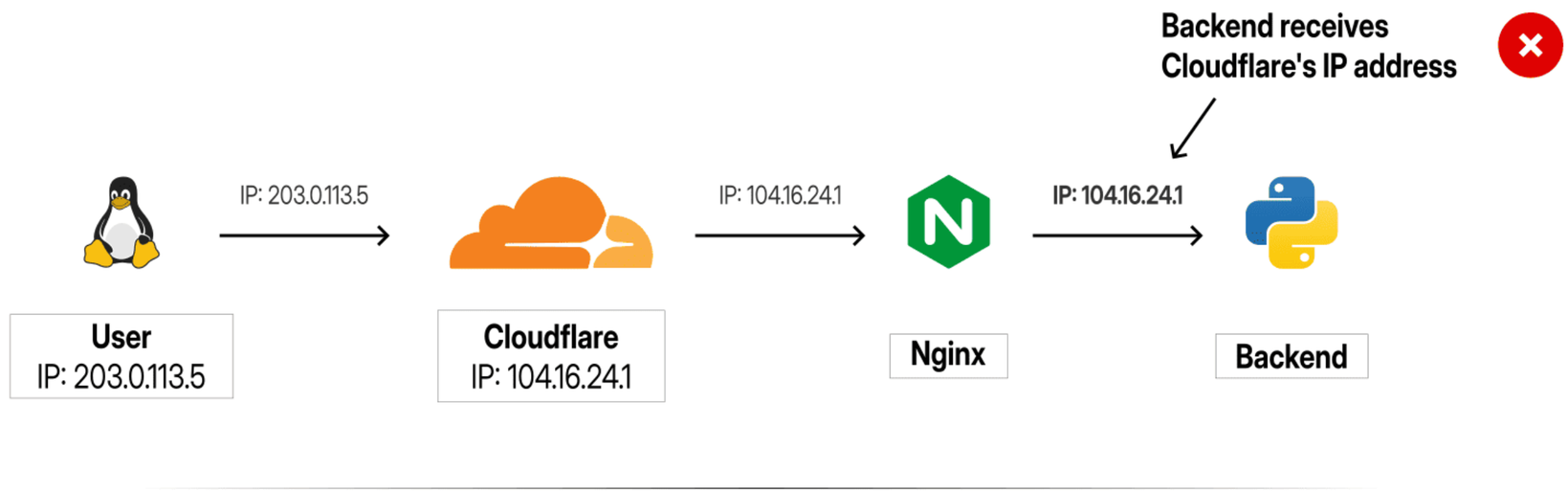
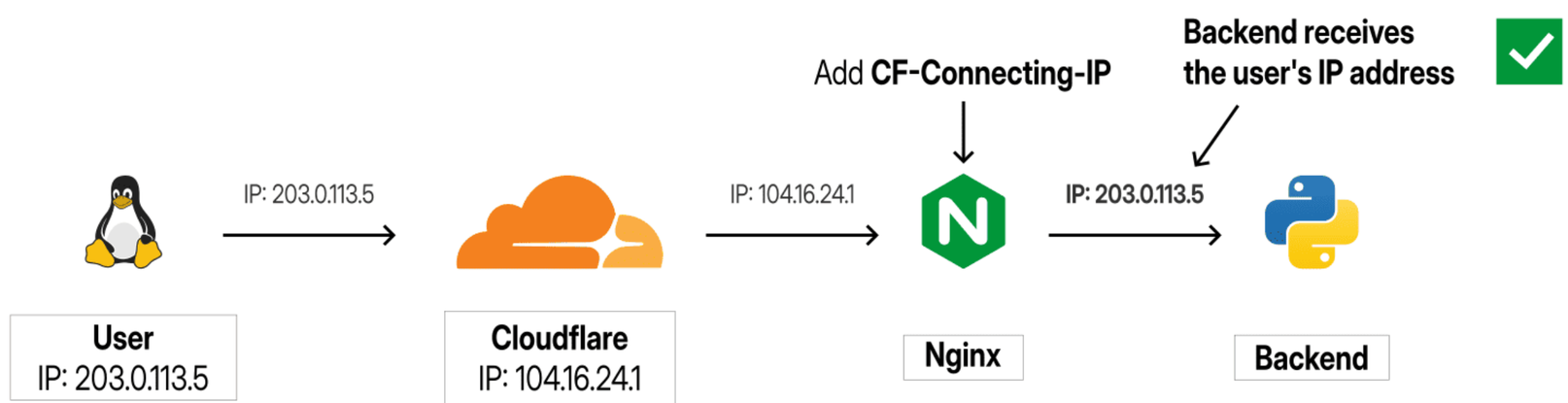




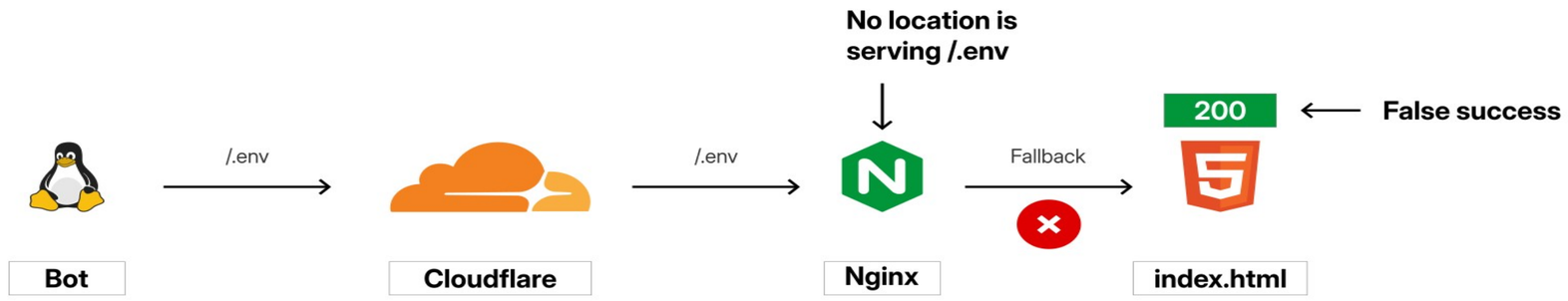
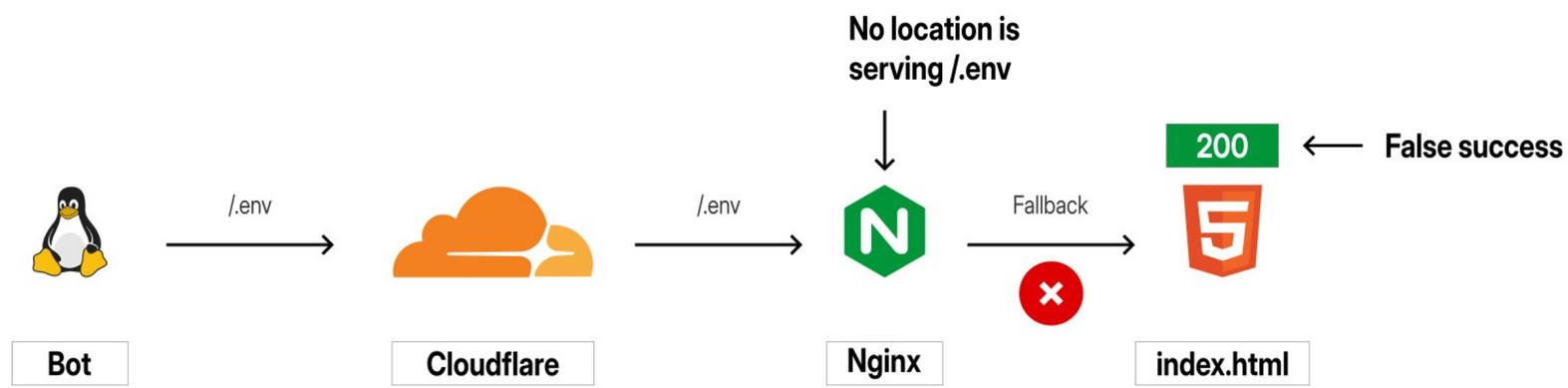
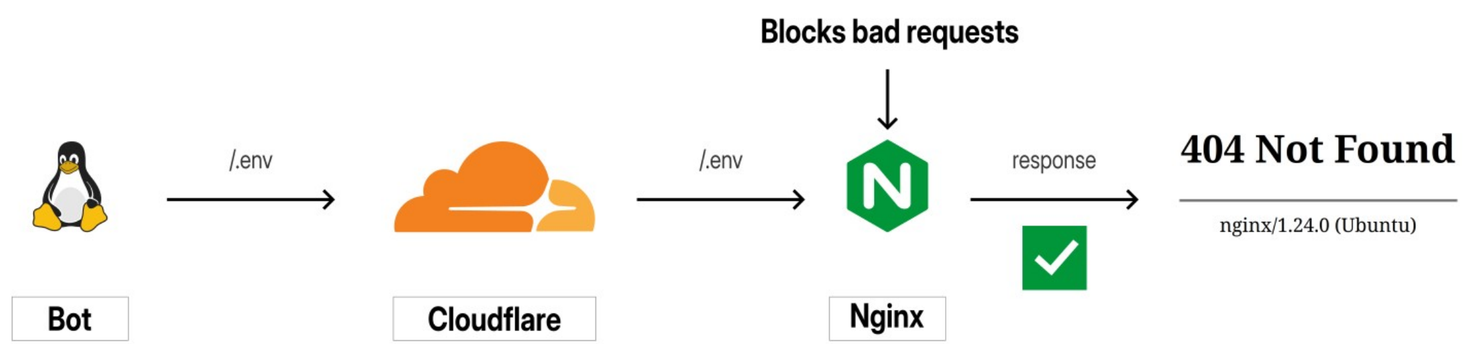
Module 4: Global delivery and security

Vulnerability scanning

Testing your defenses

- Understand and fix the **Soft 404** vulnerability.
- Configure Nginx to instantly **block malicious bots**.
- Use nmap to **scan your firewall** from the outside.

Fixing the SPA trap (Soft 404s)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